
DADS 6002 / CI 7301

Big Data Analytics

Data Ingestion Lab

Importing from MySQL to HDFS

- Let first create a simple database under MySQL.
mysql -uroot -pcloudera
mysql > create database energydata;
mysql > use energydata;
mysql > create table avgprice_by_state (
 year INT NOT NULL,

Importing from MySQL to HDFS

```
state VARCHAR(5) NOT NULL,  
sector VARCHAR(255),  
residential DECIMAL(10,2),  
industrial DECIMAL(10,2),  
transportation DECIMAL(10,2),  
other DECIMAL(10, 2),  
total DECIMAL(10,2) );  
mysql > quit;
```

Importing from MySQL to HDFS

Download a zip file from MS Teams to the shared folder and copy it to the working directory

(Source https://github.com/bbengfort/hadoop-fundamentals/raw/master/data/avgprice_kwh_state.zip)

```
# unzip avgprice_kwh_state.zip
```

Importing from MySQL to HDFS

```
# mysql -h localhost -uroot -pcloudera
--local-infile=1
mysql > use energydata;
mysql > load data local infile
'/home/cloudera/avgprice_kwh_state.csv' into table
avgprice_by_state fields terminated by ',' lines
terminated by '\n' ignore 1 lines;
mysql > quit;
```

Importing from MySQL to HDFS

```
# sqoop import --connect
jdbc:mysql://localhost:3306/energydata
--username root --password cloudera --table
avgprice_by_state --target-dir
/user/cloudera/energydata -m 1

# hadoop fs -cat /user/cloudera/energydata/part-m-
00000
```

Importing from MySQL to Hive

```
# sqoop import --connect jdbc:mysql:
//localhost:3306/energydata --username root --
password cloudera --table avgprice_by_state --
hive-table avgprice --hive-import -m 1

# hive

hive > select * from avgprice;
```

Importing from MySQL to Hbase

- Let first create a table to be imported under MySQL

```
# mysql -uroot -pcloudera
mysql > create database country_db;
mysql > use country_db;
mysql > create table country_tbl
( id int not null, country varchar(50), primary
key ( id ) );
```

Importing from MySQL to Hbase

```
mysql > insert into country_tbl values(1, 'USA' );
mysql > insert into country_tbl values(2,
'CANADA' );
mysql > insert into country_tbl values(3, 'JAPAN'
);
mysql > insert into country_tbl values(4,
'ENGLAND' );
mysql > insert into country_tbl values(5,
'THAILAND' );
mysql > select * from country_tbl;
mysql > quit;
```

Importing from MySQL to Hbase

```
# sqoop import --connect
jdbc:mysql://localhost:3306/country_db --
username root --password cloudera --table
country_tbl --hbase-table country --column-
family country-cf --hbase-row-key id --hbase-
create-table -m 1

# hbase shell
hbase > scan 'country'
```

Ingesting Product Impression Data

- Use Flume to consume the streaming user-interaction data generated by a hypothetical online store.
- Simulate an ecommerce impression log that records user interactions in the following JSON format:

```
{
  "sku" : "T9921-5"
  "timestamp" : 1453167527737
  "cid" : "51761"
  "action" : "add_cart"
  "ip" : "226.43.51.25"
}
```

Ingesting Product Impression Data

- To create the necessary directories and HDFS, download and run the script as a user with sudo privileges
- Download a flume setup shell file (flume_setup.sh) from MS Teams to the shared folder and copy it the working directory.

(Source :

<https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/setup.sh>)

Ingesting Product Impression Data

- Use nano editor to view the flume_setup.sh as follow :

```
#!/bin/bash
hadoop fs -mkdir -p /user/cloudera/impressions/
hadoop fs -chmod 777 /user/cloudera/impressions/
mkdir /tmp/impressions
chmod 777 /tmp/impressions
mkdir /tmp/flume
chmod 777 /tmp/flume
```

- Execute the setup file by
sudo su –
cd /home/cloudera
sh flume_setup.sh

Ingesting Product Impression Data

- Download a python program named impression_tracker.py from MS Teams into the shared folder and copy it to the working directory. We will execute the program to create the log file named impressions.log at /tmp/impressions.

(Source : https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/impression_tracker.py)

- Add execution privilege to the python program file then execute it as follow :
chmod +x impression_tracker.py
./impression_tracker.py

Download Examples of Configuration files

- Download the configuration files, one for client agent and one for collector agent from the MS Teams to the shared folder and then copy them to the working directory.

(Source :

<https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/client.conf>

<https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/collector.conf>)

Configure Source Agent

```
# define spooling directory source :
client.sources=r1
client.sources.r1.channels=ch1
client.sources.r1.type=spooldir
client.sources.r1.spoolDir=/tmp/impressions
# define a file channel:
client.channels=ch1
client.channels.ch1.type=FILE
```

Configure Source Agent

```
# define an Avro sink:  
client.sinks=k1  
client.sinks.k1.type=avro  
client.sinks.k1.hostname=localhost  
client.sinks.k1.port=4141  
client.sinks.k1.channel=ch1
```

Configure Collector Agent

```
# define an Avro source:  
collector.sources=r1  
collector.sources.r1.type=avro  
collector.sources.r1.bind=0.0.0.0  
collector.sources.r1.port=4141  
collector.sources.r1.channels=ch1
```

Configure Collector Agent

```
# define a file channel using multiple disks for reliability
collector.channels=ch1
collector.channels.ch1.type=FILE
collector.channels.ch1.checkpointDir=/tmp/flume/checkpoint
collector.channels.ch1.dataDir=/tmp/flume/data

# define HDFS sinks to persist events as text
collector.sinks=k1
collector.sinks.k1.type=hdfs
collector.sinks.k1.channel=ch1
```

Configure Collector Agent

```
# HDFS sink configuration
collector.sinks.k1.hdfs.path=/user/cloudera/impressions
collector.sinks.k1.hdfs.filePrefix=impressions
collector.sinks.k1.hdfs.fileSuffix=.log
collector.sinks.k1.hdfs.fileType=DataStream
collector.sinks.k1.hdfs.writeFormat=text
collector.sinks.k1.hdfs.batchSize=1000
```

Running Flume

- Open a new terminal then start a flume service.
sudo su –
service flume-ng-agent start
- Run a collector agent.
flume-ng agent --name collector --conf . --conf-file ./collector.conf
- Open a new terminal then run a collector agent.
sudo su –
flume-ng agent --name client --conf . --conf-file ./client.conf
- After finish importing, check for the imported files in the target directory /user/cloudera/impressions

Running Flume

```
# hadoop fs -ls /user/cloudera/impressions
# hadoop fs -cat
  /user/cloudera/impressions/impressions....log
to display one of the imported files
```

Kafka

- Install Kafka

```
# sudo su -
```

```
# cd /home/cloudera
```

```
# mkdir kafka
```

```
# cd kafka
```

Download Kafka from MS Teams or from the web by

```
# wget
```

```
https://archive.apache.org/dist/kafka/0.9.0.1/kafka\_2.10-0.9.0.1.tgz
```

```
# tar xzf kafka_2.10-0.9.0.1.tgz
```

Kafka

- Open a new terminal and run Kafka Server (Broker)

```
# sudo su -
```

```
# cd /home/cloudera/kafka/kafka_2.10-0.9.0.1
```

```
# bin/kafka-server-start.sh  
config/server.properties &
```

Kafka

- Open a new terminal and run a Kafka Producer

```
# sudo su -  
# cd /home/cloudera/kafka/kafka_2.10-0.9.0.1  
# bin/kafka-console-producer.sh --topic test --  
  broker-list localhost:9092
```
- Type a few lines of data to send to the Consumer.
This is a test.
Bye, Kafka.

ctrl-D to finish sending

Kafka

- Open a new terminal and run a Kafka Consumer

```
# sudo su -  
# cd /home/cloudera/kafka/kafka_2.10-0.9.0.1  
# bin/kafka-console-consumer.sh --topic test --  
  zookeeper localhost:2181 --from-beginning
```
- The consumer prints the received data from the Producer
This is a test.
Bye, Kafka.

ctrl-c to exit